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Astronauts work in a dangerous and noisy environment, and have extraordinarily compressed work schedules; they are isolated from friends and family for long periods of time. Thus they may be fatigued, depressed, or anxious. Such stressors conspire to make individual performance, and that of the team as a whole, particularly vulnerable to cognitive decrements, and it is crucial to assess cognitive ability at regular intervals while astronauts work. In a project funded by NSBRI, we have implemented a portable system, the MiniCog Rapid Assessment Battery (MRAB), which assesses 9 different cognitive functions. This system is intended to provide an "early warning alert," indicating when an astronaut or worker is suffering from stress-related deficits, such as fatigue, that may affect performance. One strength of our system is that the assessment is quick. The information can be used by the astronauts or workers themselves, either alone or as part of a team, or by a supervisor. In all cases, users can be warned to pay additional attention and take extra care, or better yet, take a break, consume food or caffeine, or even take a nap.

Task Description:	In this phase of our research, we have two major goals: First, we wish to extend MiniCog. We will reprogram the software to run on Windows Mobile PDAs in addition to Palm PDAs and to work with additional screen dimensions/resolutions. We will implement E-Prime, PsyScope, and online versions of the task battery for maximal administration flexibility. We will add the ability to enter estimates of predicted performance before and after taking a test and will compare that with actual performance; we will also add wireless relay capability to send results to a central server; and we will include an additional test that serves as an initial "screen" -- if the user has a deficit on this measure, it then will make sense to examine the precise nature of this deficit by performing the more specific tasks. Second, we will validate MiniCog. Some of these validation projects will be in collaboration with NSBRI team members David Dinges, Philip Lieberman, and Judith Orasanu. In our own lab, we will conduct three kinds of validation studies: 1) We will correlate scores on MiniCog with scores on assessment batteries currently used by NASA. 2) We will examine the effects on cognitive performance of 4 factors, specifically: a) anxiety; b) noise; c) circadian rhythm; and d) time-pressure. 3) We will examine the effects of countermeasures on mitigating the deficits produced by stressors. In particular, we will examine the effects of: a) a pre-learned short relaxation procedure; b) vigorous exercise; and, c) taking an over-the-counter stimulant (caffeine). In this past year we have extended the MiniCog system; we now have an alpha version of the MiniCog software for Windows PDAs; the Palm and Windows versions ask users to evaluate their own performance before they see their scores. We continue our validation of the MRAB, both in collaboration with the Dinges, Lieberman, and Orasanu labs, and in our own lab (the latter studies described below). Question: Is the MRAB capable of detecting the effects of restricted sleep, caffeine, and mood on very specific aspects of performance? Methods: Participants (32 female and 25 male Harvard students) practiced each of the MRAB tasks 10-12 times within a 2-week period. At the end of the 2 weeks participants came back to the lab at 5am, at which point they drank a cup of coffee (either decaf or containing 150 mg of caffeine), and then filled out personality and mood questionnaires and quietly read a prepared packet for 30 minutes. Next they completed each MRAB task once and then filled out a questionnaire asking, among other things, the amount of sleep they received the previous night and their current level of stress. Results: Data analysis is underway, but there have been interesting preliminary findings. Specifically, error rates on the 4 MRAB tasks that are sensitive to variations in attention (vigilance, filtering, divided attention, and perceptual response time) were negatively correlated (in a range of -.36 to -.50, p<=.01) with the amount of sleep participants received the previous night. Furthermore, response times on the same 4 tasks were positively correlated (in a range of .35 to .41, p<=.01) with participants' reported stress levels. Performance of the remaining 5 tasks was not related to either of these measures. Discussion: These initial analyses suggest that MRAB can detect selective effects of stress and sleep loss on specific cognitive abilities. Question: Is the MRAB capable of detecting the effects of anxiety on very specific aspects of performance? Methods: Participants (Harvard students) practice the MRAB tasks 8 times within a 2-week period. At the end of the 2-wk period participants return to the lab for a final session, during which they complete the MRAB tests and a series of mood and personality questionnaires before and after receiving instructions about a stressful public-speaking task they will perform, and then again after they have performed this task. Results: Data collection is underway. We expect that the anxiety created by the anticipation of the public speaking task and the distraction created by mental simulation of the speech both before and after its performance, will significantly interfere with performance on at least a subset of the MRAB tasks. Although public-speaking is not a particular stressor for astronauts, the anxiety that participants in this study feel, and the mental simulations they engage in, may be very similar to what is experienced by astronauts in other very high-pressure situations. We plan to continue the public-speaking study in the upcoming year, as well as to continue our software development, and begin comparing performance on MRAB in stressful conditions with performance on WinSCAT, a task battery currently in use at NASA.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The three general applications of this assessment toolbox have "earth" benefits as well as "space" benefits: First, the toolbox will provide a practical means for assessing the effects of various stressors (such as fatigue, mood, noise, drug use, or diet) on cognitive processing. We not only expect some of those variables (e.g., fatigue, certain types of drugs) to have a significant impact on astronauts, but also expect them to affect people on earth (e.g., pilots on long flights suffer from irregular sleep schedules; a large percentage of the population suffers from mood disorders or takes psychoactive medications). Second, the MRAB tasks can be used to assess the effectiveness of "countermeasures," such as training or drugs to manage fatigue. Just as the same stressors may affect people on the ground as well as the astronauts, the same countermeasures may be useful for the earthbound population as well as those in spaceflight. Finally, the tasks could be used by people "on the job" (e.g. truck drivers) to inform themselves about the current state of their cognitive processes. For example, when truck drivers "weigh in" after many hours of travel, they can run through the battery of tests and make sure that they are still mentally alert. Depending on their scores, they may want to rest before going back on the road, or they may eat a meal or drink coffee, or engage in other activities to reduce the danger of driving. Other researchers will be able to use our test battery to assess cognitive performance quickly and under a variety of laboratory and field conditions; they may also use the MiniCog platform to develop their own psychological experiments or assessment tasks. Both our test battery and the MiniCog application have an advantage over standard task batteries and many typical psychological scripting programs in that the tests are brief and the method of administration is compact, portable, and fairly inexpensive. This could make MiniCog and MRAB practical in a wide range of settings where there are questions of neurocognitive capability.

Task Progress:	We have extended the MiniCog system; we now have an alpha version of the MiniCog software for Windows PDAs; the Palm and Windows versions ask users to evaluate their own performance before they see their scores. We continue our validation of the MiniCog Rapid Assessment Battery (MRAB), both in collaboration with the Dinges, Lieberman, and Orasanu labs, and in our own lab (the latter studies described below). Question: Is the MRAB capable of detecting the effects of restricted sleep, caffeine, and mood on very specific aspects of performance? Methods: Participants (32 female and 25 male Harvard students) practiced each of the MRAB tasks 10-12 times within a 2-week period. At the end of the 2 weeks participants came back to the lab at 5am, at which point they drank a cup of coffee (either decaf or containing 150 mg of caffeine), and then filled out personality and mood questionnaires and quietly read a prepared packet for 30 minutes. Next they completed each MRAB task once and then filled out a questionnaire asking, among other things, the amount of sleep they received the previous night and their current level of stress. Results: Data analysis is underway, but there have been interesting preliminary findings. Specifically, error rates on the 4 MRAB tasks that are sensitive to variations in attention (vigilance, filtering, divided attention, and perceptual response time) were negatively correlated (in a range of -.36 to -.50, p<=.01) with the amount of sleep participants received the previous night. Furthermore, response times on the same 4 tasks were positively correlated (in a range of .35 to .41, p<=.01) with participants' reported stress levels. Performance of the remaining 5 tasks was not related to either of these measures. Discussion: These initial analyses suggest that MRAB can detect selective effects of stress and sleep loss on specific cognitive abilities. Question: Is the MRAB capable of detecting the effects of anxiety on very specific aspects of performance? Methods: Participants (Harvard students) practice the MRAB tasks 8 times within a 2-week period. At the end of the 2-wk period participants return to the lab for a final session, during which they complete the MRAB tests and a series of mood and personality questionnaires before and after receiving instructions about a stressful public-speaking task they will perform, and then again after they have performed this task. Results: Data collection is underway. We expect that the anxiety created by the anticipation of the public speaking task and the distraction created by mental simulation of the speech both before and after its performance, will significantly interfere with performance on at least a subset of the MRAB tasks. Although public-speaking is not a particular stressor for astronauts, the anxiety that participants in this study feel, and the mental simulations they engage in, may be very similar to what is experienced by astronauts in other very high-pressure situations. We plan to continue the public-speaking study in the upcoming year, as well as to continue our software development, and begin comparing performance on MRAB in stressful conditions with performance on WinSCAT, a task battery currently in use at NASA.
Bibliography Type:	Description: (Last Updated: 01/31/2008)
Articles in Peer-reviewed Journals	Shephard JM, Kosslyn SM. "The MiniCog Rapid Assessment Battery: Developing a “blood-pressure cuff for the mind“." Aviat Space Environ Med. 2005 Jun;76(6 Suppl):B192-7. PMID: 15943212 , Jun-2005